Lesson 8: Linking environment and history

90 minutes / introductory course / working draft

Reading evidence from tree rings

Observation means an inspected feature such as a ring width or scar. Dating places a ring in a calendar year through crossdating and anchoring. Interpretation concerns events such as felling, use, disturbance or climate change. A secure ring date does not automatically determine when timber was felled or used.

Archaeologists examine contexts and documents; ecologists compare trees and scars; climatologists calibrate and test against climate observations. Before combining sources, check differences in time, location and season.

Case: wooden containers and dated slips

A study of Nara containers compares ring dates with written dates on wooden slips. Distinguish the event recorded on a slip from the year of the last preserved ring. Examine association, preservation and the sequence of use and deposition.

The Nara National Research Institute for Cultural Properties publishes ring-width values, images and measurement paths. A paper and a public dataset do not necessarily contain the same specimens. Check specimen identifiers and archaeological context before linking them. Access routes appear at the end of this handout.

Exercise A: three timber dates

All examples are fictional. A document refers to a building in 1210. Samples were taken from different parts of the present structure. Spend 8 minutes on supported claims, limits and further evidence.

Timber A: last preserved ring 1187, bark absent. Timber B: last preserved ring 1199, bark present. Timber C: last preserved ring 1213, bark absent.

1. What does each year represent? What can be said about felling and use?

2. How could each timber relate to the 1210 record?

3. How could repair, reuse, storage, the meaning of the record and dating error be compared?

4. What should be checked about position, joints, working traces, bark and outer wood?

Hints

The number of outer rings lost from A and C is unknown, so the last preserved ring alone cannot determine the felling year. C formed a ring in 1213, prompting a reassessment of its relationship to the 1210 record. Later repair or additions are possibilities to evaluate using structure and context.

B formed a ring in 1199. Even with bark, inspect the outer ring and position within the ring before deciding felling season; the interval before use also remains to be considered. Timber dates alone do not fix a building's construction or repair date.

Exercise B: a research plan or literature review

Spend 8 minutes: draft for 5, review for 2 and revise for 1. Choose one format and state the question, sources, comparison and limits.

Research plan

Possible opening: “Were the timbers at a site associated with the building's construction or a later repair?” Specify the material, sampling and preservation, dating method and comparison evidence. What positions or working traces would be expected under a repair hypothesis? Compare another explanation. Identify alternatives if sampling is not possible.

This is an illustrative proposal, not a report of actual materials or findings. Do not present planned measurements or expected results as completed observations.

Literature review

Possible opening: “How can the relationship between temperature and ring width change over time?” Find relevant papers and compare region, species, season, calibration and verification periods, results and limits. Do not count articles based on the same data as independent evidence. Consider differences in samples and methods when results disagree.

Peer review

Do the sources address the question? Is there a competing explanation? What remains unresolved and what evidence could change the conclusion? Are figure sources, units and scope clear?

Report and routes to evidence

Choose a research plan or a review of studies or cases. Write by hand during the final 45 minutes of Lesson 8 and submit in class. You may consult course materials and notes. There is no minimum page or reference count, and figures are optional. Link the question, materials and method, comparison, evidence, conclusion and limits. Identify any sources you use as clearly as you can. The report alone determines the course grade.

Finding papers and data

Follow a DOI to the publisher's article page and check the authors, year, title, methods and figures. Full text is not always freely available. Check a library or an author-hosted copy; if needed, politely request a copy from the corresponding author. For practical steps see Supplement BG06: accessing research papers.

• Containers and wooden slips: Maeda et al. (2024) • Nara institute: archaeological tree-ring dataset announcement • Hinoki and early spring temperature: Yonenobu and Eckstein (2006) • Northeast Asia spring temperature: Ohyama et al. (2013) • International Tree-Ring Data Bank: NOAA

Identify the source and scope of figures and data. Distinguish measurements, synthetic data and conceptual diagrams. Check the conditions for using external images and collaborative research materials.

The current final report requires one question in about four paragraphs within 45 minutes, including planning and checking. No new literature search is required. Use the course case materials; further searching is optional study after class.

Lesson 8 Develop one question in 45 minutes

Write and submit in the final 45 minutes of the class. The final report is the sole basis of grading; classroom participation and learning behaviour are not graded.

Scope

Choose one case discussed in class, or a specific subject of your own. Write either a research proposal or a case review in about four paragraphs. New experiments, measurements or literature searches are not required.

Paragraph	Content
1	Define one question, subject, region and period.
2	Explain which tree-ring evidence and methods address it.
3	Discuss one alternative explanation and evidence for comparison.
4	State the limits of the conclusion and what to check next.

A practical guide is about 250–400 English words. This is not a minimum or a grading criterion. A shorter report can meet the criteria if its evidence and reasoning are sufficient. Use a short title that states your question in the task field of the supplied paper.

Course materials and notes are allowed; internet access is unnecessary. A reference such as “Lesson 5, slide 7” is sufficient to identify a course source. There is no required number of references or figures. Distinguish proposed work from completed measurements.

Suggested time allocation

Five minutes to choose a question and evidence; 30 minutes to write; ten minutes to check reasoning, alternatives, limits and identification details. Timing choices are not graded.

Lesson 8 A short example and the rubric

This fictional building and its dates illustrate reasoning. This is not a required answer to copy; build your own question and evidence.

1. I ask how far timber can constrain a building date. In this fictional practice case, the last surviving ring of a beam is assigned to 1750, while a documentary record concerns 1758. Neither bark nor sapwood survives. Are the ring date and construction date interchangeable?

2. I would first inspect support for the 1750 assignment: the securely dated reference, overlap, matching features and investigation of mismatches. The exercise provides no crossdating record, so 1750 is a stated assumption, not a result I have independently established.

3. Lesson 5 explains why loss of the outer wood matters. The last surviving ring does not alone fix the felling year; later rings may have been removed. Even if the document concerns construction, it may not apply to this beam. Later repair or reuse of older timber are alternative explanations.

4. Neither felling nor original construction is established by these facts alone. I would examine the original document and the part of the building it describes, joints and working traces, and other timbers retaining their outer edges. Comparing their dates could help distinguish a single construction episode from repair or reuse. Source for the principle: Lesson 5, slides 5–8. The building and dates are fictional practice assumptions.

Criterion (20 points each)	Where it appears in the example
Question and scope	1: subject and distinction between dates
Evidence and methods	2: what must support the crossdate
Comparison and logic	3: loss of wood, repair, reuse and documentary context
Evidence and referencing	3–4: source location and separation of fictional assumptions
Conclusion, limits and next steps	4: limits and specific additional evidence

Total: 100 points. Use 20, 15, 10 or 5 as attainment guides for each criterion; no relevant statement receives 0. Length or format alone does not determine a full score; judge the reasoning and evidence actually presented.